You arrange the LED light-output to vary with time, and detect this using the voltage across a light-dependent resistor. However, this voltage is small and needs to be amplified. Your teacher suggests that you use a linear amplifier.

Question 9

Explain in words what is meant by the term **linear** in the context of the amplifier.

2 marks

Question 10

On the set of axes provided in Figure 7, sketch a graph of the output voltage, V_{OUT}, as a function of the input voltage V_{IN}. The axes of the graph must be clearly labelled.

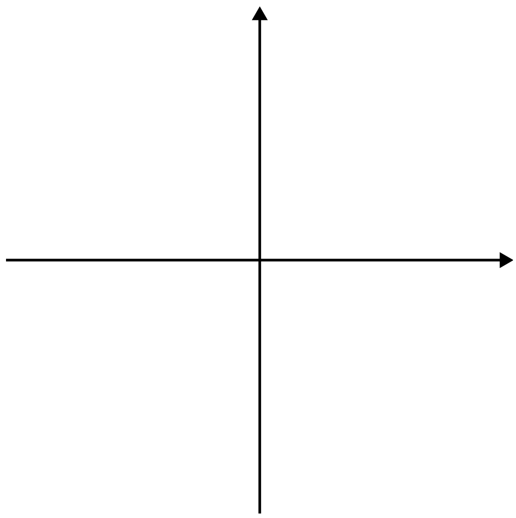

Figure 7

2 marks

A friend of yours has a wine collection and asks you to help design a way of keeping a record of the collection using the bar codes printed on the wine bottle labels. Each wine type has a unique bar code. Your teacher shows you how to combine the LED circuit and detector circuit to form a bar code reader and you design the necessary digital electronics.

A typical bar code is shown in Figure 8a. It is essentially a set of white and black bands that can be represented by binary states: white = 0 and black = 1. To simplify this problem you may assume that there are only 4 bands in the bar code. An example, and the associated binary code (1010), when read by a bar code reader, is shown in Figure 8b.

Figure 8a

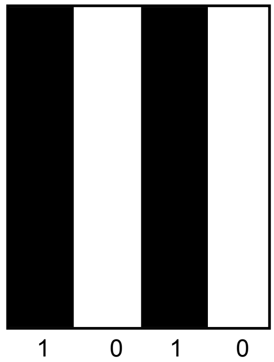

Figure 8b

Question 11

Your friend decides to have a different bar code for each type of wine. What is the maximum number of wine types that they can have in their collection?

Explain your answer.

____

____

____

____

____

Number of wine types =

2 marks

To identify a bottle as it is added or removed from the wine collection, you design a system where the code read from the wine bottle is compared to codes stored in a computer memory. Each bit of the bottle bar code is compared to the appropriate bit of the stored codes till a match is found. Each bit comparison produces a 1 if the bits match and a 0 if they do not.

Question 12

Which one of the following truth tables correctly provides this output? In the top row of these truth tables, B represents the bottle bar-code bit, C the stored computer code bit and O the output bit.

A.

B	C	O
0	0	1
0	1	0
1	0	0
1	1	0

B.

B	C	O
0	0	1
0	1	0
1	0	0
1	1	1

C.

B	C	O
0	0	0
0	1	1
1	0	1
1	1	0

D.

B	C	O
0	0	0
0	1	0
1	0	0
1	1	1

2 marks

In addition to identifying the wine type, you suggest a circuit that will count the bottles. This counting circuit uses three JK flip-flops (JKFF) as shown in Figure 9. The input to the first JKFF is also shown when a new wine bottle is identified to be counted. It is the input change on the edge numbers $1 \rightarrow 0$, that causes a change to the output, Q, of the JKFF.

There are initially 3 bottles, and this is indicated by the JKFF outputs, Q, in Figure 9. Note that the JKFF outputs are read in reverse order as indicated.

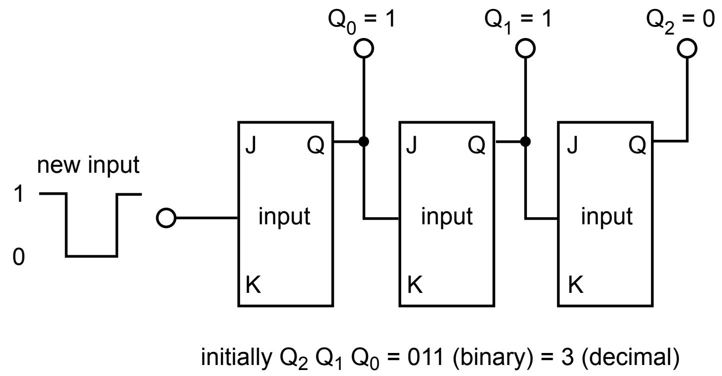

Figure 9

Question 13

By completing the timing diagram below (Figure 10), show how the outputs of the JKFFs change to indicate there is a new total of 4 bottles.

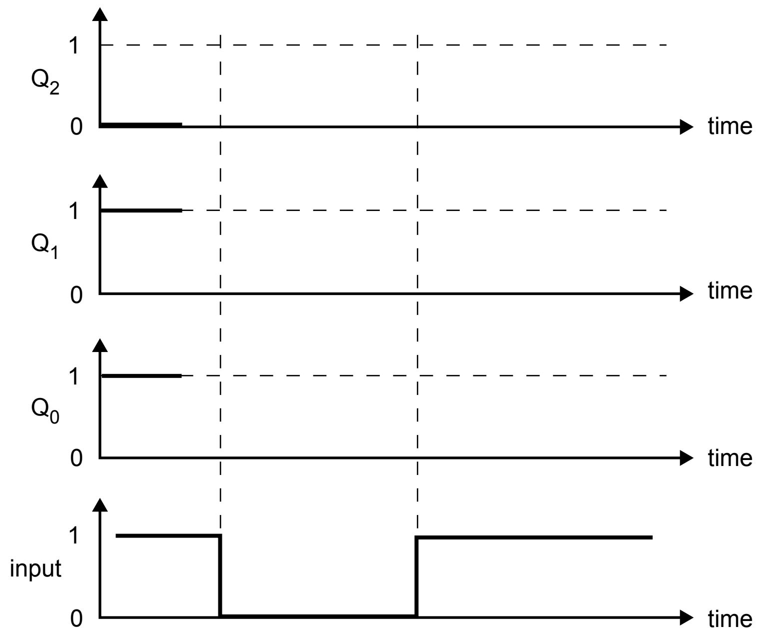

Figure 10

3 marks

END OF QUESTION AND ANSWER BOOK